In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```
df = pd.read_csv("academic_performance_dataset_2.csv")
```

In [4]:
```
print("Missing values per column:\n")
print(df.isnull().sum())
```

```
Missing values per column:

Student_ID       0
Age              2
Gender           0
Study_Hours      0
Attendance       0
Math_Score       0
Science_Score    0
English_Score    1
Final_Grade      0
dtype: int64
```

In [5]:
```
numeric_cols = ["Age", "Attendance", "Math_Score", "Science_Score", "English_Sco

for col in numeric_cols:
    df[col] = df[col].fillna(df[col].mean())
```

In [6]:
```
df["Gender"] = df["Gender"].fillna(df["Gender"].mode()[0])
```

In [7]:
```
df["Gender"] = df["Gender"].replace({
    "Male": "M",
    "Female": "F"
})
```

In [8]:
```
print("Missing values after cleaning:\n")
print(df.isnull().sum())

print("\nUnique values in Gender:\n", df["Gender"].unique())
```

```
Missing values after cleaning:

Student_ID       0
Age              0
Gender           0
Study_Hours      0
Attendance       0
Math_Score       0
Science_Score    0
English_Score    0
Final_Grade      0
```

```
dtype: int64

Unique values in Gender:
 ['M' 'F']
```

In [9]:

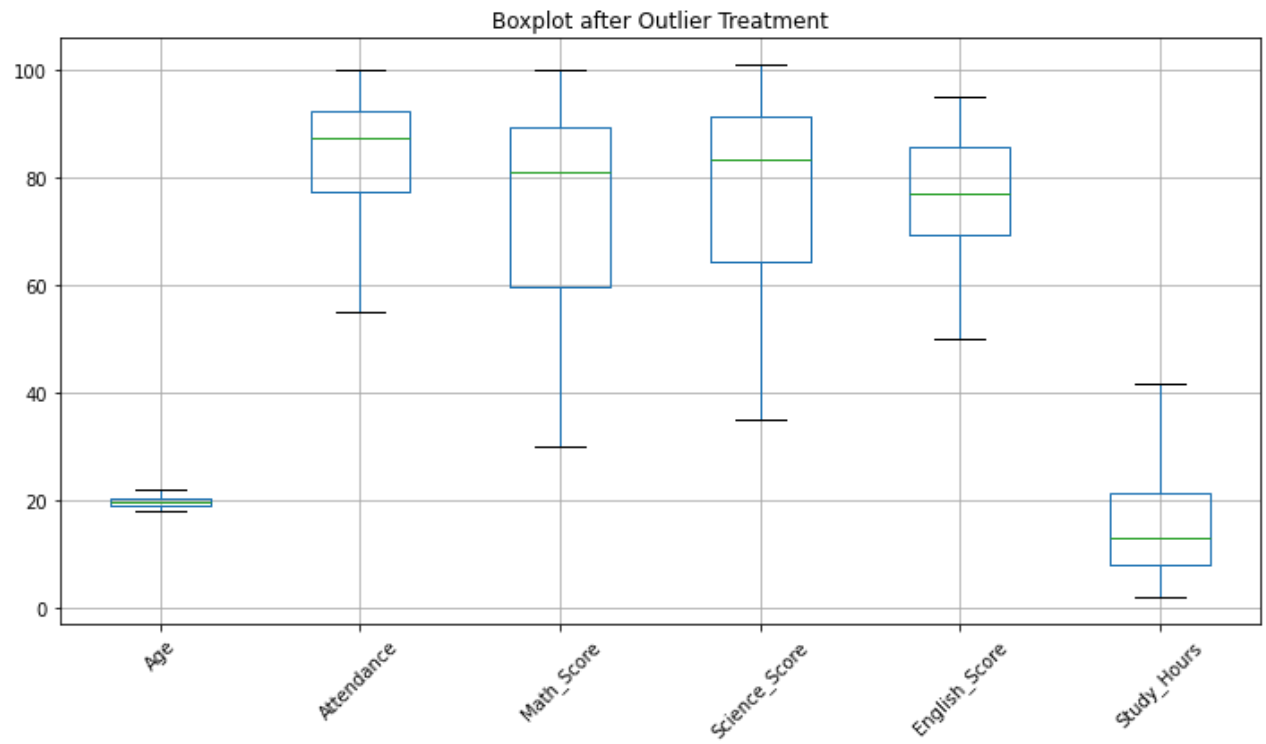
```
def detect_outliers_iqr(column):
    Q1 = column.quantile(0.25)
    Q3 = column.quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    return lower_bound, upper_bound
```

In [10]:

```
for col in numeric_cols:
    lb, ub = detect_outliers_iqr(df[col])
    df[col] = np.clip(df[col], lb, ub)
```

In [11]:

```
plt.figure(figsize=(12,6))
df[numeric_cols].boxplot()
plt.title("Boxplot after Outlier Treatment")
plt.xticks(rotation=45)
plt.show()
```



In [12]:

```
print("Skewness before transformation:")
print(df["Study_Hours"].skew())
```

```
Skewness before transformation:
1.0100569939656718
```

In [13]:

```
df["Study_Hours_Log"] = np.log1p(df["Study_Hours"])
```

In [14]:

```
print("\nSkewness after transformation:")
print(df["Study_Hours_Log"].skew())
```

```
Skewness after transformation:
-0.22689097493086258
```

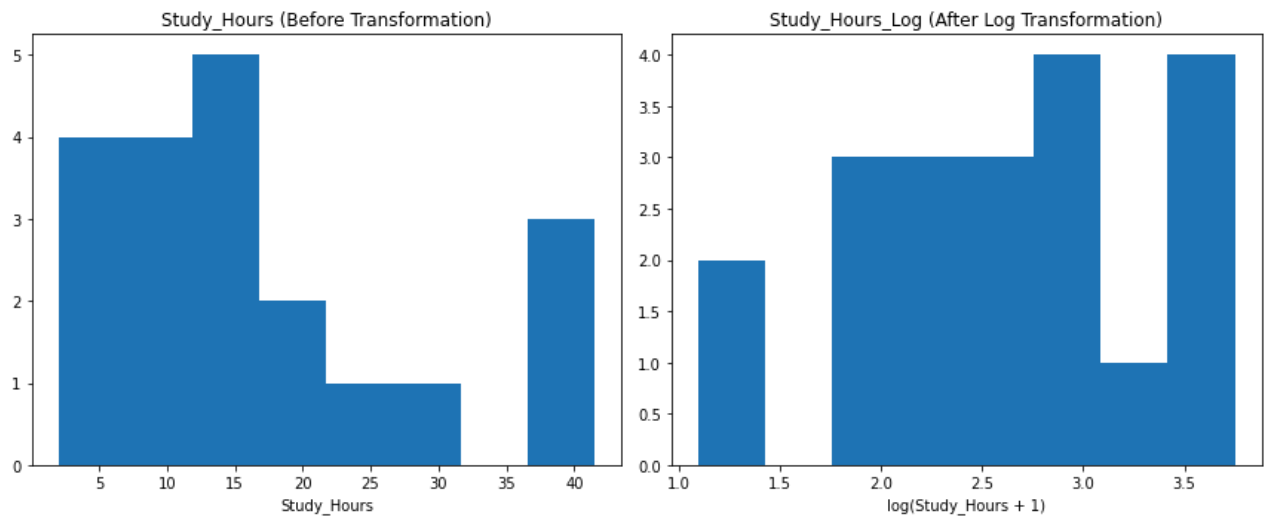
In [15]:

```
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.hist(df["Study_Hours"], bins=8)
plt.title("Study_Hours (Before Transformation)")
plt.xlabel("Study_Hours")

plt.subplot(1,2,2)
plt.hist(df["Study_Hours_Log"], bins=8)
plt.title("Study_Hours_Log (After Log Transformation)")
plt.xlabel("log(Study_Hours + 1)")

plt.tight_layout()
plt.show()
```



In [16]:

```
df
```

Out[16]:

	Student_ID	Age	Gender	Study_Hours	Attendance	Math_Score	Science_Score	English_Sc
0	1	18.000000	M	10.0	85	78	80	75.000
1	2	19.000000	F	12.0	90	85	88	82.000
2	3	20.000000	M	8.0	78	60	65	70.000
3	4	18.000000	M	15.0	88	90	92	88.000
4	5	21.000000	F	20.0	95	95	96	90.000
5	6	19.000000	F	25.0	92	88	90	76.368
6	7	19.666667	M	5.0	70	55	60	65.000
7	8	20.000000	F	30.0	98	98	99	92.000
8	9	22.000000	M	3.0	65	45	50	55.000

	Student_ID	Age	Gender	Study_Hours	Attendance	Math_Score	Science_Score	English_Sc
9	10	19.000000	F	18.0	89	82	85	80.000
10	11	18.000000	M	12.0	80	75	78	73.000
11	12	21.000000	F	40.0	100	100	101	95.000
12	13	20.000000	M	6.0	75	58	63	68.000
13	14	19.000000	M	7.0	82	62	67	72.000
14	15	19.666667	F	41.5	60	30	35	50.000
15	16	22.000000	F	9.0	93	89	91	85.000
16	17	18.000000	M	14.0	87	80	82	78.000
17	18	20.000000	F	16.0	90	85	87	83.000
18	19	21.000000	M	2.0	55	40	42	60.000
19	20	19.000000	F	41.5	99	97	98	90.000

In [17]:

```
from scipy.stats import zscore
```

In [18]:

```
z_scores = df[numeric_cols].apply(zscore)

z_scores.head()
```

Out[18]:

	Age	Attendance	Math_Score	Science_Score	English_Score	Study_Hours
0	-1.360828	0.115120	0.168523	0.134257	-0.112552	-0.549740
1	-0.544331	0.512086	0.515484	0.555457	0.463193	-0.386854
2	0.272166	-0.440632	-0.723660	-0.655492	-0.523798	-0.712626
3	-1.360828	0.353300	0.763312	0.766057	0.956688	-0.142525
4	1.088662	0.909052	1.011141	0.976657	1.121187	0.264689

In [19]:

```
outliers = (np.abs(z_scores) > 3)

print("Outlier count per column (Z-score method):\n")
print(outliers.sum())
```

```
Outlier count per column (Z-score method):

Age              0
Attendance       0
Math_Score       0
Science_Score    0
English_Score    0
Study_Hours      0
dtype: int64
```

In [20]:

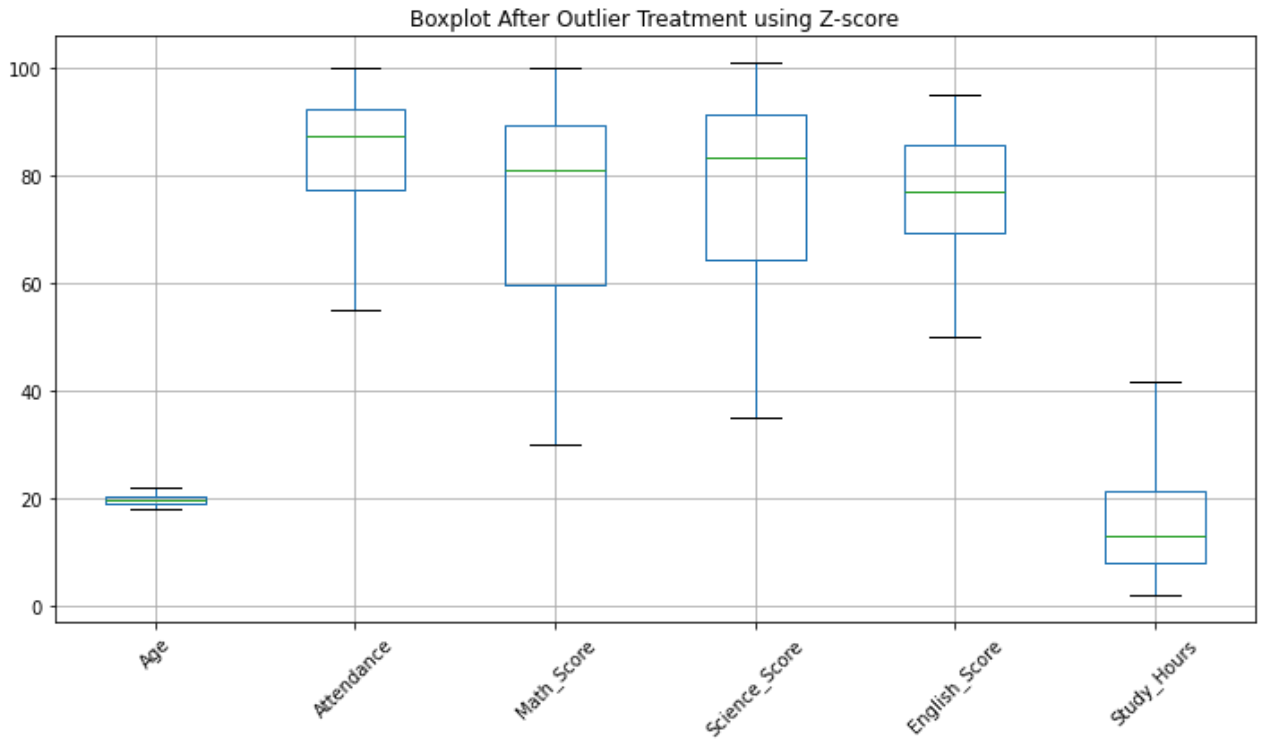
```
df_zscore = df.copy()

for col in numeric_cols:
```

```
    median_val = df_zscore[col].median()
    df_zscore.loc[np.abs(zscore(df_zscore[col])) > 3, col] = median_val
```

In [21]:

```
plt.figure(figsize=(12,6))
df_zscore[numeric_cols].boxplot()
plt.title("Boxplot After Outlier Treatment using Z-score")
plt.xticks(rotation=45)
plt.show()
```



In [22]:

```
df_zscore
```

Out[22]:

	Student_ID	Age	Gender	Study_Hours	Attendance	Math_Score	Science_Score	English_Sc
0	1	18.000000	M	10.0	85.0	78	80.0	75.000
1	2	19.000000	F	12.0	90.0	85	88.0	82.000
2	3	20.000000	M	8.0	78.0	60	65.0	70.000
3	4	18.000000	M	15.0	88.0	90	92.0	88.000
4	5	21.000000	F	20.0	95.0	95	96.0	90.000
5	6	19.000000	F	25.0	92.0	88	90.0	76.368
6	7	19.666667	M	5.0	70.0	55	60.0	65.000
7	8	20.000000	F	30.0	98.0	98	99.0	92.000
8	9	22.000000	M	3.0	65.0	45	50.0	55.000
9	10	19.000000	F	18.0	89.0	82	85.0	80.000
10	11	18.000000	M	12.0	80.0	75	78.0	73.000
11	12	21.000000	F	40.0	100.0	100	101.0	95.000
12	13	20.000000	M	6.0	75.0	58	63.0	68.000

	Student_ID	Age	Gender	Study_Hours	Attendance	Math_Score	Science_Score	English_Sc
13	14	19.000000	M	7.0	82.0	62	67.0	72.000
14	15	19.666667	F	41.5	60.0	30	35.0	50.000
15	16	22.000000	F	9.0	93.0	89	91.0	85.000
16	17	18.000000	M	14.0	87.0	80	82.0	78.000
17	18	20.000000	F	16.0	90.0	85	87.0	83.000
18	19	21.000000	M	2.0	55.0	40	42.0	60.000
19	20	19.000000	F	41.5	99.0	97	98.0	90.000

In [ ]: